AI-Powered Financial Fraud Detection System

Summer Training Project 2026

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The Fraud Landscape (2025-2026)

- **The Sophistication Shift:** Financial fraud is evolving rapidly, with a recent 180% increase in advanced, coordinated attack tactics.
- **AI-Generated Threats:** Attackers are increasingly leveraging deepfakes, synthetic identities, and sophisticated automation to bypass standard verification.
- **Scale & Speed:** Modern financial ecosystems require dynamic defenses, as static rule-based systems simply cannot keep pace with real-time, novel fraud vectors.

Limitations vs Innovation

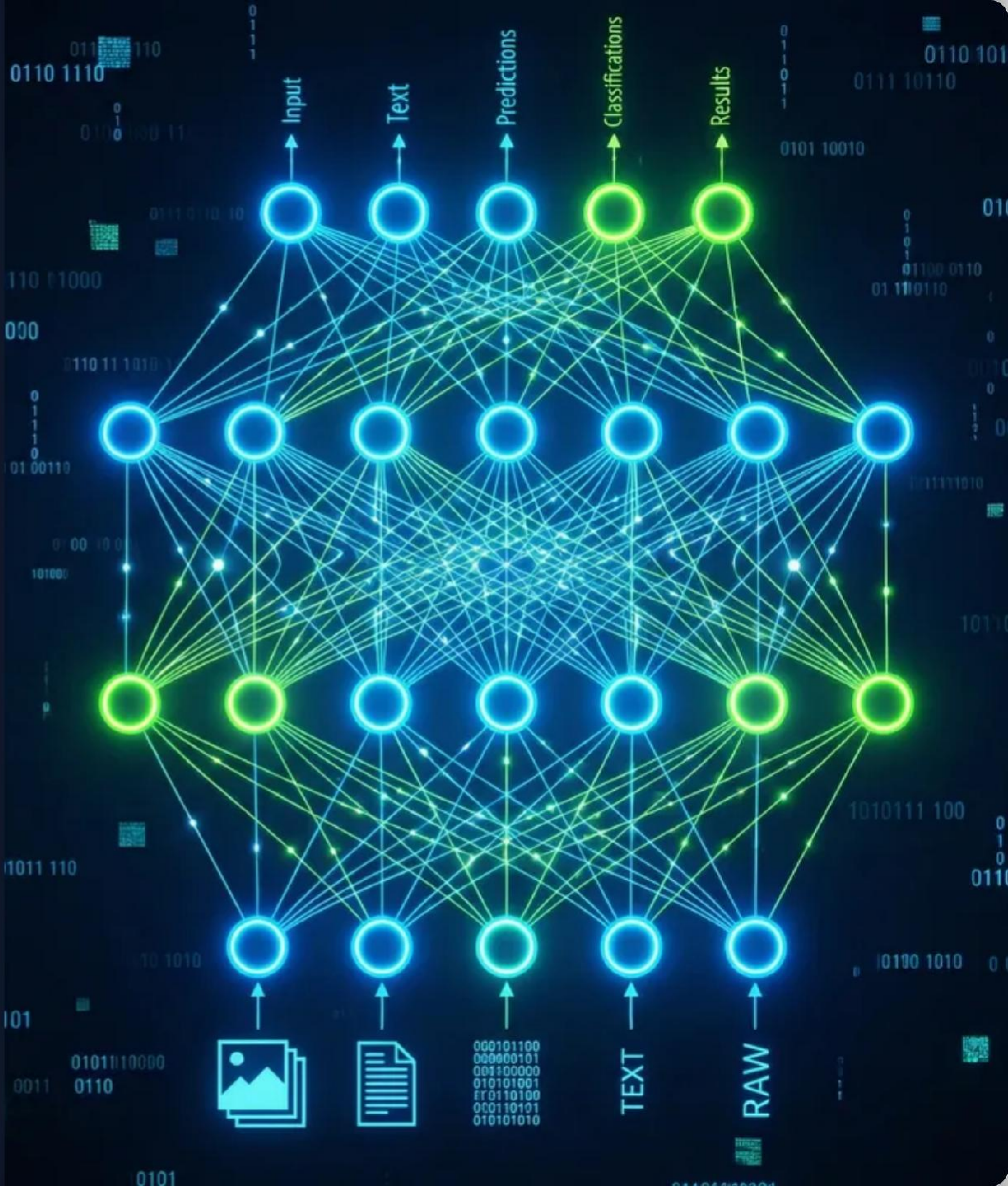
- **Rule-Based Systems:** Rigid "if-then" logic results in high false positive rates. They are reactive and cannot detect unknown or zero-day patterns without manual updates.
- **Manual Review:** Costly, slow, and unscalable. By the time human analysts investigate a new vector, significant financial losses have already occurred.
- **AI-Driven Defense:** Analyzes hundreds of variables simultaneously, learning from historical patterns to adapt to emerging threats in milliseconds.

System **Architecture**

Our proposed pipeline ingests raw transaction data—including amounts, merchant categories, geolocation, and timestamps—in real-time via high-throughput streams.

The data undergoes rigorous automated feature engineering before being evaluated by an ensemble of machine learning models.

This hybrid architecture guarantees sub-millisecond decision latency, accurately blocking malicious actors without disrupting legitimate customer checkouts.



Core Machine Learning Models

Random Forest Classifier

Supervised Learning Engine

Acts as the primary decision engine. By utilizing an ensemble of distinct decision trees, it inherently reduces variance and handles the complex, non-linear nature of transactional data robustly. It requires labeled historical data to train effectively.

Isolation Forest

Unsupervised Anomaly Detection

Perfect for identifying rare, hidden anomalies in large datasets without any prior labeling. It acts as a safety net against novel, zero-day fraud tactics that the supervised models have not yet encountered.

Performance Benchmark (F1-Score)

Random Forest provides the optimal balance. While Neural Networks score slightly higher, Random Forest maintains the interpretability and explainability strictly required for regulatory compliance in the financial sector.

Tools & Libraries

Python 3.x

The core programming language chosen for its extensive ecosystem, readability, and industry-standard status in data science and machine learning.

Scikit-Learn

The primary ML library used for data preprocessing (StandardScaler), model building (Random Forest Classifier), and generating evaluation metrics.

Pandas

Employed for robust data manipulation, structural formatting, and handling complex tabular datasets before feeding them into the ML models.

NumPy

Utilized under the hood for highly optimized, multidimensional numerical computations and rapid array processing across the pipeline.

Implementation Details

Data Preprocessing

Handled missing values, normalized numerical features using StandardScaler, and encoded categorical variables to prepare the dataset for inference.

Class Imbalance Mitigation

Applied class weighting techniques dynamically within the model to ensure it doesn't become biased towards the majority (legitimate) class.

Evaluation Metrics

Shifted focus away from raw accuracy towards ROC-AUC and F1-Score, which are the true indicators of success for highly skewed fraud datasets.

Data Generation

Utilized specialized algorithms to securely synthesize realistic transaction data, preserving complete privacy while ensuring robust model training.

Questions?

Thank you for your time and attention.

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Image Sources

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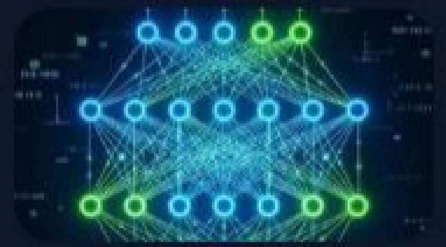

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